The trinity theorem

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Abstract

In this paper, I describe the trinity theorem and prove it.
The paper ends with "The End"

The trinity theorem

Theorem 1. *For reals x, y and z, with $s = x + y + z$,*

$$2(x+y+z)\,e^{x+y+z} = \sum_{n=1}^{\infty}\sum_{r=0}^{\infty}\sum_{k=0}^{n} \frac{\binom{n+r-1}{r}\binom{n}{k}}{(n+r-1)!}\Big(x^r y^k z^{n-k} + y^r z^k x^{n-k} + z^r x^k y^{n-k}\Big),$$

where e is the base of the natural logarithm,

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!},$$

and $0! = 1$ and, for $n > 0$, $n! = n\,(n-1)!$.

Remark. The index n starts at 1, because for $n = 0$ and $r = 0$ the factor $(n+r-1)! = (-1)!$ is undefined, while for $n = 0$ and $r \geq 1$ we have $\binom{r-1}{r} = 0$, so no term is lost. The index k runs from 0 to n, since z^{n-k} requires $n - k \geq 0$ and $\binom{n}{k} = 0$ for $k > n$.

Proof

Absolute convergence. Replacing x, y, z by $|x|, |y|, |z|$ does not decrease any term, and the computation below shows that the resulting series converges. Hence the series converges absolutely, and its terms may be summed in any order.

Step 1: the k-sum. By the binomial theorem,

$$\sum_{k=0}^{n}\binom{n}{k} y^k z^{n-k} = (y+z)^n,$$

and similarly $\sum_{k=0}^{n}\binom{n}{k} z^k x^{n-k} = (z+x)^n$ and $\sum_{k=0}^{n}\binom{n}{k} x^k y^{n-k} = (x+y)^n$. Hence the triple sum equals $S_x + S_y + S_z$, where

$$S_x = \sum_{n=1}^{\infty}\sum_{r=0}^{\infty} \frac{\binom{n+r-1}{r}}{(n+r-1)!}\, x^r (y+z)^n,$$

and S_y, S_z are obtained from S_x by cyclically permuting $x \to y \to z \to x$.

Step 2: the coefficient. Put $m = n + r - 1 \geq 0$. Then

$$\frac{\binom{m}{r}}{m!} = \frac{1}{m!}\cdot\frac{m!}{r!\,(m-r)!} = \frac{1}{r!\,(n-1)!}.$$

Step 3: the double sum factors. Let $a = y + z$. Then

$$S_x = \sum_{r=0}^{\infty}\frac{x^r}{r!}\sum_{n=1}^{\infty}\frac{a^n}{(n-1)!} = e^x \cdot a \sum_{j=0}^{\infty}\frac{a^j}{j!} = e^x\, a\, e^a = (y+z)\,e^{x+y+z}.$$

Step 4: summation. By symmetry, $S_y = (z+x)\,e^{x+y+z}$ and $S_z = (x+y)\,e^{x+y+z}$. Therefore

$$S_x + S_y + S_z = \big[(y+z) + (z+x) + (x+y)\big]e^{x+y+z} = 2(x+y+z)\,e^{x+y+z}. \qquad \blacksquare$$

Remark. The three terms of the theorem are cyclic copies of the single identity

$$\sum_{r=0}^{\infty}\sum_{n=1}^{\infty}\frac{x^r a^n}{r!\,(n-1)!} = a\,e^{x+a}.$$

The End